

Example

. A beam ABC of length 9 m has one support at the left end and the other support at a distance of 6 m from the left end. The beam carries a point load of 12 kN at right end and also carries a uniformly distributed load of 4 kN/m over a length of 3 m as shown in Fig. Determine the slope and deflection at point C.

Take $E = 2 \times 10^5 \text{ N/mm}^2$ and $I = 5 \times 10^8 \text{ mm}^4$.

Sol. Given :

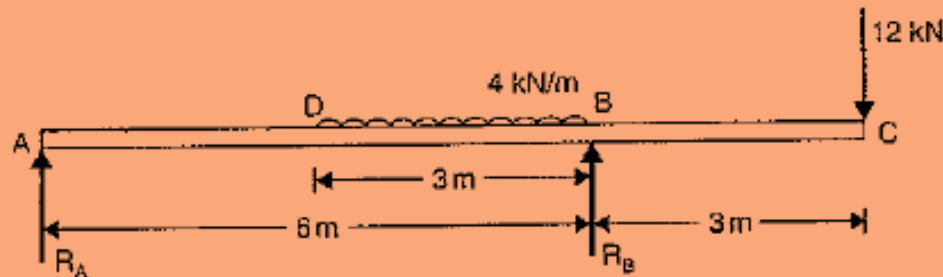
Point load, $W = 12 \text{ kN}$

U.d.l., $w = 4 \text{ kN/m}$

Value of $E = 2 \times 10^5 \text{ N/mm}^2$

Value of $I = 5 \times 10^8 \text{ mm}^4$

First calculate the reactions R_A and R_B .



Taking moments about A, we get

$$\begin{aligned}R_B \times 6 &= 4 \times 3 \times \left(3 + \frac{3}{2}\right) + 12 \times 9 \\&= 54 + 108 = 162\end{aligned}$$

$$\therefore R_B = \frac{162}{6} = 27 \text{ kN } (\uparrow)$$

and

$$R_A = \text{Total load} - R_B = 24 - 27 = -3 \text{ kN } (\downarrow)$$

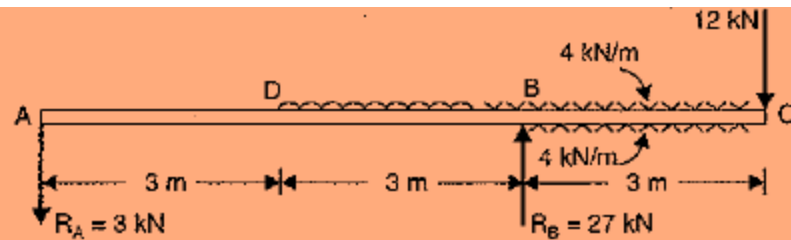


Fig. 12.14

The B.M. at any section at a distance x from the support A is given by,

$$EI \frac{d^2 y}{dx^2} = -R_A \times x \quad \vdots \quad -4(x-3) \frac{(x-3)}{2} \quad \vdots \quad +R_B(x-6) \quad \vdots \quad +4(x-6) \frac{(x-6)}{2}$$

$$= -3x \quad \vdots \quad -2(x-3)^2 \quad \vdots \quad +27(x-6) \quad \vdots \quad +2(x-6)^2$$

Integrating the above equation, we get

$$EI \frac{dy}{dx} = -\frac{3x^2}{2} + C_1 \quad \vdots \quad -\frac{2(x-3)^3}{3} \quad \vdots \quad +\frac{27(x-6)^2}{2} + \frac{2(x-6)^3}{3} \quad \dots(i)$$

Integrating again, we get

$$EI y = -\frac{3x^3}{2 \cdot 3} + C_1 x + C_2 \quad \vdots \quad -\frac{2(x-3)^4}{3 \cdot 4} \quad \vdots \quad +\frac{27(x-6)^3}{2 \cdot 3} \quad \vdots \quad +\frac{2(x-6)^4}{3 \cdot 4}$$

or

$$EI y = -\frac{x^3}{2} + C_1 x + C_2 \quad \vdots \quad -\frac{(x-3)^4}{6} \quad \vdots \quad +\frac{9}{2}(x-6)^3 \quad \vdots \quad +\frac{1}{6}(x-6)^4 \quad \dots(ii)$$

where C_1 and C_2 are constant of integration. Their values are obtained from boundary conditions which are :

(i) at $x = 0, y = 0$ and

(ii) at $x = 6 \text{ m}, y = 0$.

(i) Substituting the $x = 0$ and $y = 0$ in equation (ii) upto first dotted line (as $x = 0$ lies in the first part AD of the beam), we get

$$0 = 0 + C_1 \times 0 + C_2 \quad \therefore C_2 = 0$$

(ii) Substituting $x = 6$ and $y = 0$ in equation (ii) upto second dotted line (as $x = 6$ lies in the second part DB of the beam), we get

$$\begin{aligned} 0 &= -\frac{6^3}{2} + C_1 \times 6 + 0 - \frac{(6-3)^4}{6} \\ &= -108 + 6C_1 - 13.5 = -121.5 + 6C_1 \end{aligned}$$

or
$$C_1 = \frac{121.5}{6} = 20.25$$

Substituting the values of C_1 and C_2 in equations (i) and (ii), we get

$$EI \frac{dy}{dx} = \frac{3}{2} x^2 + 20.25 \left[\begin{array}{c} \vdots \\ \vdots \end{array} \right] - \frac{2(x-3)^3}{3} \left[\begin{array}{c} \vdots \\ \vdots \end{array} \right] + \frac{27}{2} (x-6)^2 \left[\begin{array}{c} \vdots \\ \vdots \end{array} \right] + \frac{2}{3} (x-6)^3 \dots (iii)$$

and
$$EIy = -\frac{x^3}{2} + 20.2 \times x \left[\begin{array}{c} \vdots \\ \vdots \end{array} \right] \left| \begin{array}{c} \vdots \\ \vdots \end{array} \right| - \frac{1}{6}(x-3)^4 \left[\begin{array}{c} \vdots \\ \vdots \end{array} \right] \left| \begin{array}{c} \vdots \\ \vdots \end{array} \right| + \frac{9}{2} (x-6)^3 \left[\begin{array}{c} \vdots \\ \vdots \end{array} \right] + \frac{1}{6} (x-6)^4 \dots (iv)$$

(a) *Slope at the point C*

By substituting $x = 9$ m in equation (iii), we get the slope at C. Here complete equation is to be taken as point $x = 9$ m lies in the last part of the beam.

$$\begin{aligned}\therefore EI.\theta_C &= -\frac{3}{2} \times 9^2 + 20.25 - \frac{2}{3} (9-3)^3 + \frac{27}{2} (9-6)^2 + \frac{2}{3} (9-6)^3 \\ &\quad \left(\because \frac{dy}{dx} \text{ at } C = \theta_C \right) \\ &= -121.5 + 20.25 - 144 + 121.5 + 18 = -105.75 \text{ kNm}^2 \\ &= -105.75 \times 10^3 \times 10^6 \text{ Nmm}^2 = -105.75 \times 10^9 \text{ Nmm}^2 \\ \therefore \theta_C &= -\frac{105.75 \times 10^9}{2 \times 10^5 \times 5 \times 10^8} = -0.0010575 \text{ radians. Ans.}\end{aligned}$$

(b) *Deflection at the point C*

By substituting $x = 9$ m in complete equation (iv), we get the deflection at C.

$$\begin{aligned}\therefore EI \times y_C &= -\frac{9^3}{2} + 20.25 \times 9 - \frac{1}{6} (9-3)^4 + \frac{9}{2} (9-6)^3 + \frac{1}{6} (9-6)^4 \\ &= -364.5 + 182.25 - 216 + 121.5 + 13.5 \\ &= -263.25 \text{ kNm}^3 = -263.25 \times 10^{12} \text{ Nmm}^3 \\ \therefore y_C &= -\frac{263.25 \times 10^{12}}{2 \times 10^5 \times 5 \times 10^8} = -2.6325 \text{ mm. Ans.}\end{aligned}$$

Chapter 5

Shear Stress in Beams

The average shear stress = shear force / Area

The maximum shear stress = 1.5 times the average shear stress.

The *shear stress* in a beam at any transverse cross-section in its length, and at a point a vertical distance y from the neutral axis, *resulting from bending* is given by

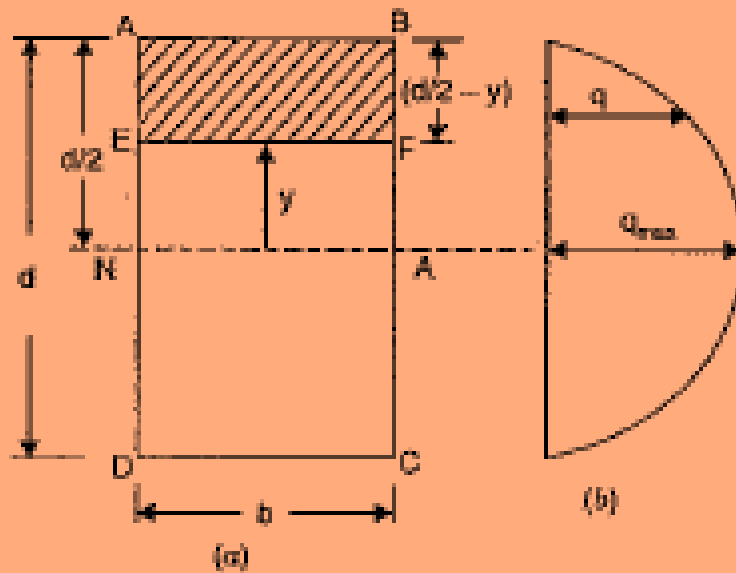
$$\tau = \frac{QA\bar{y}}{Ib}$$

where Q is the applied vertical shear force at that section; A is the area of cross-section “above” y , i.e. the area between y and the outside of the section, which may be above or below the neutral axis (N.A.); $\bar{y}$ is the distance of the centroid of area A from the N.A.; I is the second moment of area of the complete cross-section; and b is the breadth of the section at position y .

Shear stress distribution for different cross sections

For rectangular sections,

$$\tau = \frac{6Q}{bd^3} \left[\frac{d^2}{4} - y^2 \right] \quad \text{with} \quad \tau_{\max} = \frac{3Q}{2bd} \quad \text{when} \quad y = 0$$



For **I-section beams** the *vertical shear* in the web is given by

$$\tau = \frac{Q}{2I} \left[\frac{h^2}{4} - y^2 \right] + \frac{Qb}{2It} \left[\frac{d^2}{4} - \frac{h^2}{4} \right]$$

with a *maximum* value of

$$\tau_{\max} = \frac{Qh^2}{8I} + \frac{Qb}{2It} \left[\frac{d^2}{4} - \frac{h^2}{4} \right]$$

The *maximum* value of the *horizontal shear* in the flanges is

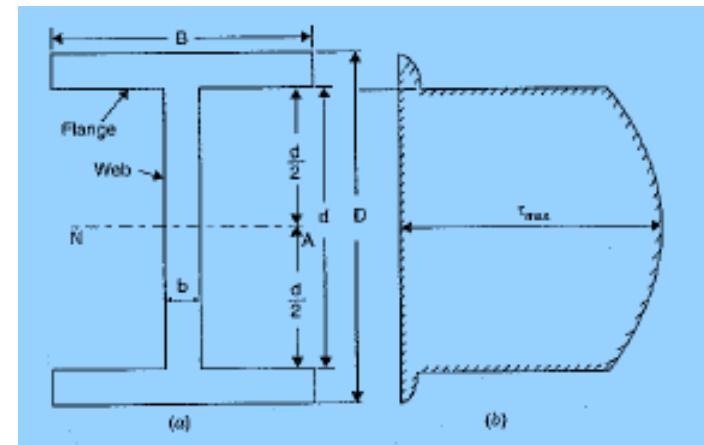
$$\tau_{\max} = \frac{Qb}{4I} (d - t_1)$$

For **circular sections**

$$\tau = \frac{4Q}{3\pi R^2} \left[1 - \left(\frac{y}{R} \right)^2 \right]$$

with a *maximum* value of

$$\tau_{\max} = \frac{4Q}{3\pi R^2}$$



Problem 8.2. A beam of cross-section of an isosceles triangle is subjected to a shear force of 30 kN at a section where base width = 150 mm and height = 450 mm. Determine :

- (i) horizontal shear stress at the neutral axis,
- (ii) the distance from the top of the beam where shear stress is maximum, and
- (iii) value of maximum shear stress.

Sol. Given :

Shear force at the section, $F = 30 \text{ kN} = 30,000 \text{ N}$

Base width, $CD = 150 \text{ mm}$

Height, $h = 450 \text{ mm}$.

(i) Horizontal shear stress at the neutral axis

The neutral axis of the triangle is at a distance of $\frac{h}{3}$ from base or $\frac{2h}{3}$ from the apex B . Hence distance of neutral axis from B will be $\frac{2 \times 450}{3} = 300 \text{ mm}$ as shown in Fig. 8.3 (a). The width of the section at neutral axis is obtained from similar triangles BCD and BNA as

$$\frac{NA}{CD} = \frac{300}{450}$$

or
$$NA = \frac{300}{450} \times CD = \frac{300}{450} \times 150 = 100 \text{ mm}.$$

The shear stress at any section is given by equation (8.1) as

$$\tau = F \times \frac{A \times \bar{y}}{I \times b} \quad \dots(i)$$

where τ = Shear stress at the section

F = Shear force = 30,000 N

A = Area above the axis at which shear stress is to be obtained
[i.e., shaded area of Fig. 8.3 (a)]

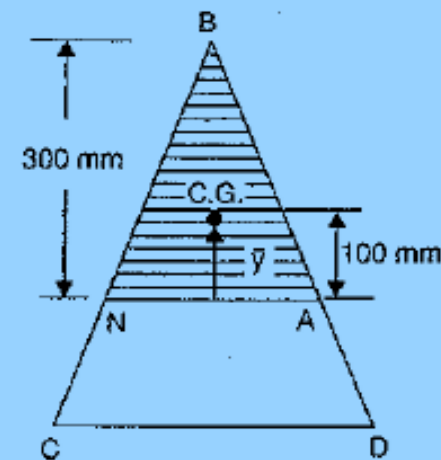


Fig. 8.3 (a)

$$= \frac{NA \times 300}{2} = \frac{100 \times 300}{2} = 15000 \text{ mm}^2$$

$\bar{y}$ = Distance of the C.G. of the area A from neutral axis

$$= \frac{1}{3} \times 300 = 100 \text{ mm}$$

I = M.O.I. of the total section about neutral axis

$$= \frac{\text{Base width} \times \text{Height}^3}{36} \quad \left(\because \frac{B \times h^3}{36} \text{ where } B = \text{Base Width of Triangle} \right)$$

$$= \frac{150 \times 450^3}{36} \text{ mm}^4$$

b = Actual width of the section at which shear stress is to be obtained

$$= NA = 100 \text{ mm.}$$

Substituting these values in equation (i), we get

$$\begin{aligned} \tau &= 30,000 \times \frac{15000 \times 100}{\left(\frac{150 \times 450}{36} \right)^3 \times 100} \text{ N/mm}^2 \\ &= 1.185 \text{ N/mm}^2. \quad \text{Ans.} \end{aligned}$$

(ii) The distance from the top of the beam where shear stress is maximum

Let the shear stress is maximum at the section EF at a distance x from the top of the beam as shown in Fig. 8.3 (b). The distance EF is obtained from similar triangles BEF and BCD as

$$\frac{EF}{CD} = \frac{x}{450}$$

$$\therefore EF = \frac{x}{450} \times CD = \frac{x}{450} \times 150 = \frac{x}{3}$$

The shear stress at the section EF is given by equation (8.1) as

$$\tau = F \times \frac{A \times \bar{y}}{I \times b} \quad \dots(ii)$$

where $F = 30,000 \text{ N}$

A = Area of section above EF i.e., Area of shaded triangle BEF

$$\begin{aligned} &= \frac{EF \times x}{2} = \frac{x}{3} \times \frac{x}{2} & \left(\because EF = \frac{x}{3} \right) \\ &= \frac{x^2}{6} \end{aligned}$$

$\bar{y}$ = Distance of C.G. of the Area A from neutral axis

$$= \frac{2h}{3} - \frac{2x}{3} = \frac{2 \times 450}{3} - \frac{2x}{3} = \left(300 - \frac{2x}{3} \right)$$

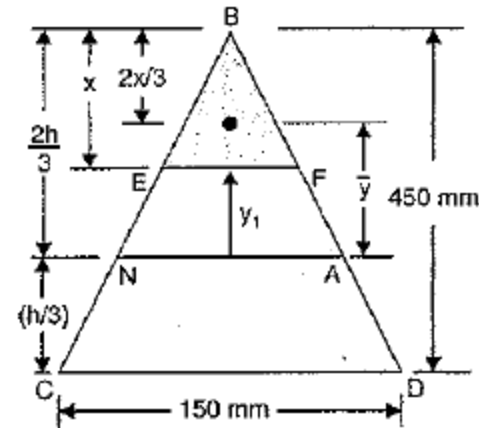


Fig. 8.3 (b)

$I = \text{M.O.I. of } \triangle BCD \text{ about neutral axis}$

$$= \frac{150 \times 450^3}{36} \text{ mm}^4$$

$$b = \text{Width of section } EF = \frac{x}{3}.$$

Substituting these values in equation (ii), we get

$$\begin{aligned}\tau &= \frac{30,000 \times \left(\frac{x^2}{6}\right) \times \left(300 - \frac{2x}{3}\right)}{\left(\frac{150 \times 450^3}{36}\right) \times \frac{x}{3}} = 0.0000395x \left(300 - \frac{2x}{3}\right) \\ &= 0.0000395 \left(300x - \frac{2x^2}{3}\right) \quad \dots(iii)\end{aligned}$$

For maximum shear stress $\frac{d\tau}{dx} = 0$

$$\text{or} \quad 300 - \frac{2}{3} \times 2x = 0 \quad \text{or} \quad 300 = \frac{4x}{3}$$

$$\text{or} \quad x = \frac{300 \times 3}{4} = 225 \text{ mm. Ans.}$$

Hence, shear stress is maximum at a distance of 225 mm from the top of the beam.

(iii) *Value of Maximum Shear Stress*

The value of maximum shear stress will be obtained by substituting $x = 225$ mm in equation (iii).

$$\begin{aligned}\therefore \text{Maximum shear stress} &= 0.0000395 \left(300 \times 225 - \frac{2}{3} \times 225^2\right) \\ &= 1.333 \text{ N/mm}^2. \text{ Ans.}\end{aligned}$$

Problem 8.4. A timber beam of rectangular section is simply supported at the ends and carries a point load at the centre of the beam. The maximum bending stress is 12 N/mm^2 and maximum shearing stress is 1 N/mm^2 , find the ratio of the span to the depth.

Sol. Given :

Maximum bending stress, $\sigma_{max} = 12 \text{ N/mm}^2$

Maximum shear stress, $\tau_{max} = 1 \text{ N/mm}^2$.

Let b = Width of the beam,

d = Depth of the beam,

L = Span of the beam,

W = Point load at the centre.

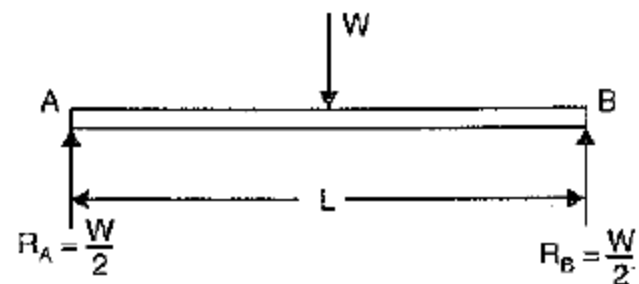


Fig. 8.5

Maximum shear force, $F = \frac{W}{2}$

and maximum B.M., $M = \frac{W \times L}{4}$

Now average shear stress, $\tau_{avg} = \frac{\text{Shear force}}{\text{Area}} = \frac{\left(\frac{W}{2}\right)}{b \times d} = \frac{W}{2bd}$.

Maximum shear stress is given by equation (8.4)

$$\therefore \tau_{max} = 1.5 \times \tau_{avg}$$

or

$$1 = 1.5 \times \frac{W}{2bd}$$

$$\left(\because \tau_{max} = 1, \tau_{avg} = \frac{W}{2bd} \right)$$

or

$$\frac{W}{bd} = \frac{2}{1.5} = \frac{4}{3} \quad \dots(i)$$

Now using bending equation

$$\frac{M}{I} = \frac{\sigma}{y} \quad \text{or} \quad \sigma = \frac{M}{I} \times y$$

$\therefore$ Maximum bending stress,

$$\sigma_{max} = \frac{M}{I} \times y_{max}$$

$$= \frac{\frac{W \times L}{4} \times \frac{d}{2}}{\frac{bd^3}{12}}$$

$$\left(\because y_{max} = \frac{d}{2} \right)$$

$$= \frac{12}{8} \times \frac{W.L.d}{bd^3} = 1.5 \frac{W.L}{bd^2}$$

or

$$12 = 1.5 \frac{W.L}{bd^2}$$

$$(\because \sigma_{max} = 12)$$

$$= 1.5 \frac{W}{bd} \cdot \frac{L}{d}$$

$$= 1.5 \times \frac{4}{3} \times \frac{L}{d}$$

$$= 2 \times \frac{L}{d}$$

$$\frac{L}{d} = \frac{12}{2} = 6. \text{ Ans.}$$

$$\left[\because \frac{W}{bd} = \frac{4}{3} \text{ from equation (i)} \right]$$